

DIGPALSINH RATHOD

Process Engineer

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Summary

Process Engineer with 3+ years of experience in pharmaceutical manufacturing and chemical process development. Proven expertise in process optimization, scale-up, tech transfer, and quality compliance (cGMP, FDA, WHO). Adept at root cause analysis, safety process, validation protocols, and cross-functional collaboration to improve product yield and reduce downtime. Strong foundation in chemical engineering with hands-on knowledge of process simulation tools and regulatory documentation.

Work Experience

Process Engineer, Pfizer, USA

Dec 2024 – Present

- Led process improvement initiatives for sterile injectable and oral solid dosage products, reducing batch cycle times by 18% and improving throughput by 25% using Lean Six Sigma tools.
- Conducted root cause analysis on deviations and out-of-specification (OOS) results, implementing CAPAs that improved product quality and compliance across 3 production lines.
- Supported scale-up and technology transfer from R&D to manufacturing for new product launches, aligning protocols with cGMP and FDA expectations.
- Designed and implemented robust process validation protocols (IQ/OQ/PQ), ensuring reproducibility and regulatory approval for high-volume products.
- Collaborated with cross-functional teams (QA, QC, R&D, Engineering) to ensure seamless product lifecycle management from development to commercial scale.
- Led investigations on process deviations, performed risk assessments (FMEA), and ensured documentation in TrackWise and SAP for audit readiness.
- Utilized Aspen Plus and ChemCAD to simulate and optimize chemical process parameters, improving solvent recovery and energy efficiency by 15%.
- Trained junior engineers and technicians on process control, safety protocols, and Good Documentation Practices (GDP).

Process Engineer, Sun Pharmaceutical Industries Ltd, India 2023

July 2021 – Aug

- Executed process development, scale-up, and optimization of Active Pharmaceutical Ingredients (API), leading to 12% cost savings via solvent reuse and process redesign.
- Prepared process flow diagrams (PFD), P&IDs, and SOPs for chemical operations including crystallization, filtration, distillation, and drying.
- Oversaw pilot plant trials and batch production, optimizing reaction conditions to enhance yield by 20% while maintaining compliance with cGMP.
- Conducted process risk assessments (HAZOP, What-If) and implemented safety controls, significantly reducing incident rates.
- Supported regulatory filings by preparing process narratives, batch manufacturing records, and validation protocols for USFDA, EMA, and WHO audits.
- Reviewed and approved change controls, deviation reports, and conducted root cause analysis using 5 Why/Fishbone methodology.
- Partnered with R&D to scale up 10+ new APIs, ensuring smooth tech transfer from lab to commercial production.
- Improved equipment cleaning procedures by developing robust CIP (Clean-in-Place) strategies, reducing turnaround time by 30%.
- Maintained thorough documentation in SAP and LIMS, supporting traceability and process transparency.

Certifications

- Basic of Disaster Risk Management (GIDM)
- Natural Gas engineering (NPTEL-IIT, Guwahati)
- Six sigma: Green belt (LinkedIn-learning)

Technical & Core Skills

- Process Engineering:** Process scale-up & optimization, Tech transfer (R&D to plant), Batch & continuous manufacturing, Equipment qualification (IQ/OQ/PQ), PFDs & P&ID creation, CIP/SIP systems
- Quality & Compliance:** cGMP, ICH, FDA, WHO compliance, Deviation & CAPA management, RCA, 5 Whys, Fishbone, HAZOP, FMEA, QbD & PAT, Audit readiness
- Tools & Simulation:** Aspen Plus, HYSYS, ChemCAD, Minitab, JMP, MATLAB, SPC, DOE, Six Sigma tools, SCADA, DCS, PLC
- Chemical Operations:** Crystallization, filtration, distillation, API & formulation processes, Solvent recovery, Mass & heat transfer
- Systems & Documentation:** SAP, MES, LIMS, TrackWise, Documentum, eBMR, GDP, ALCOA+
- Soft Skills:** Cross-functional teamwork, Problem solving, Project coordination, Technical writing, Lean mindset

Education

Master in Chemical Engineering
Rowan University, USA

Sept 2023 – Dec 2024

Bachelor in Chemical Engineering
2022
GSFC University, India

Aug 2018 – May